

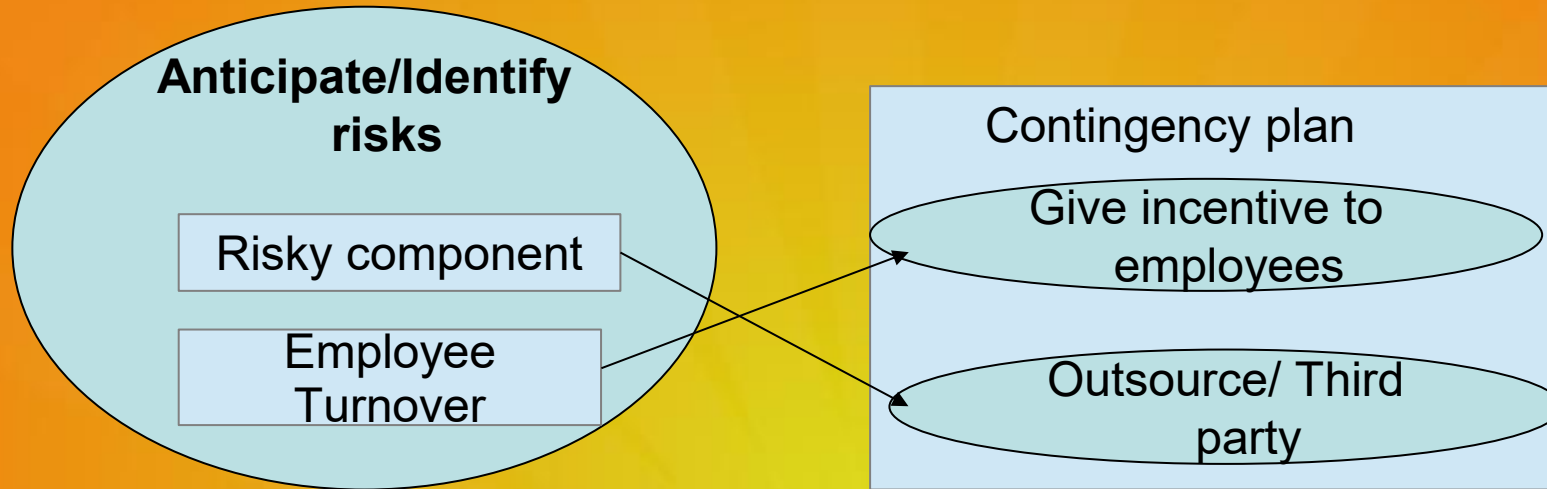
Risk management

What is risk?

- Risks are **potential problems** that might affect the successful completion of a software project.
- Risk is an **uncertain event or condition** that, if it occurs has a **positive negative effects** on project's objectives.
- Risk is a potential problem it might happen or might not.
- But regardless of the outcome it is good idea to identify it assess its Probability of occurrence, estimate its impact and establish contingency Plan.

When risk turns to reality
it hampers successful and timely completion of project

- Basic characteristics of risk:
- *Uncertainty*: Risk may or may not happen
- *Loss*: If risk becomes a reality losses will occur



Risk management

- Risk analysis and management are intended to help a software team understand and manage uncertainty during the development process.
- Recognizing what can go wrong is the first step called **risk identification**.
- Next each risk is analyzed to determine the **likelihood** that it will occur and the damage that it will do if it does occur.
- Once this information is established, risk are ranked **,by probability and impact**.
- Finally, a plan is developed to manage those risk that have high probability and high impact.

Risk strategies

- They are of two types:

Reactive risk strategy "What went wrong?"

- Monitors the project for likely risks
- Resources are set aside to deal with risks when they become actual problem
- Software team does nothing about risks until something goes wrong.

Then ,the team attempts to correct the problem ,
this is often called a fire fighting mode.

- When this fails, “crisis management” takes over and the project is
in real jeopardy(failure).

e.g., patching software after a hack

Proactive risk strategy:

"What could go wrong?"

- . Better than reactive risk strategy
- . Begins long before technical work is initiated.
- . Potential risks are identified, their probability and impacts are assessed and ranked by importance.
- . Software team establishes a plan for managing risks.
- . As all the risks can not be avoided, the team works to develop a contingency (plan that will enable it to respond in a controlled and effective manner)

Examples include **proactive** safety training, hazard inspections, etc

Types of Risk

- If the risk becomes a reality, unwanted consequences or losses will occur.
- When risks are analyzed it is important to quantify the level of uncertainty and the degree of loss associated with each risk.
- To accomplish this, different categories of risk are considered.

1. Project risks

(Affecting schedule, budget, resources (e.g., scope creep, resource shortage).

- Threaten the project plan. i.e if project risks become real it is likely that the project schedule will slip and costs will increases.
- Risk factors:
 - ▣ Potential budgetary
 - ▣ Schedule
 - ▣ Personnel(staffing and organization)
 - ▣ Resource
 - ▣ Stakeholder
 - ▣ Requirements problem

Technical risks

(Related to technology, design, performance, or implementation (e.g., new tech challenges, integration issues))

- Threaten the quality and timeliness of the software to be produced.
- If a technical risks becomes a reality implementation may become difficult or impossible.
- Risk factors:
 - ▣ Potential design
 - ▣ Implementation
 - ▣ Interface
 - ▣ Verification
 - ▣ And maintenance problems
 - ▣ In addition specification ambiguity, technical uncertainty, technical obsolescence and leading-edge technology are also risk factors

Technical risk occur because the problem is harder to solve than
we
Thought it would be.

Business risk

(Affecting market viability, organizational goals (e.g., market changes, user adoption))

- Threaten the viability of the software to be produced and even the project or the product.
- Top five business risk are:
 1. **Market risk** :building an excellent product or system that no one really wants
 2. **Strategic risk** :building a product that no longer fits into the overall business strategy for the company
 3. **Sales risk** :building a product that the sales force doesn't understand how to sell.
 4. **Management risk** :losing the support of senior management due to change in focus or a change in people.
 5. **Budget risk** :losing budgetary or personnel commitment.

Known risk

- Those that can be uncovered after careful evaluation of the project plan, the business and technical environment in which the project is being developed ,and other reliable information source
- e.g. unrealistic delivery date ,lack of documented requirements ,poor development environment.

Predictable/unpredictable risk

Predictable risk: Risks identified through historical data, past projects, or current analysis.

Examples:

Budget overruns due to past project cost escalations.

Staffing shortages (e.g., known high turnover in a department).

Website crashes during peak sales (predictable traffic spikes).

Management: Can be proactively managed, modeled, and planned against using data analytics and established procedures.

Unpredictable risks: Risks that are extremely difficult to identify in advance .

Examples: A novel pandemic.

Sudden geopolitical shifts.

Risk identification

- Systematic attempt to specify threats to the project plan.
- Predictable and known risks can be avoided and controlled possibly by identification.
- There are two types of risks for each categorized risk:
- **Generic risks** : potential threat to every software project.(universal threats like poor requirements, resource issues)
- **Product-specific risks** : can be identified only by those with a clear understanding of technology ,the people and the environment specific for building the project. (unique challenges tied to a project's tech, team, or environment, like new tech integration or customer behavior)

- One method for identifying risks is to create a risk item checklist. The checklist focuses on some subset:
- Product size
- Business impact
- Stakeholders characteristics
- Process identification
- Development environment
- Technology to be build
- Staff size and experience
- After organizing risk item checklist ,a set of “risk components and drivers ”are listed along with their probability of occurrence ,helps in risk identification.

Risk component and driver(factors)

- Risk component:
- Performance : degree of uncertainty that **the product will meet its requirements** and be fit for its intended use.
- Cost : degree of uncertainty that the project budget will be maintained. eg. "Can we stay within budget?"
- Schedule : degree of uncertainty that the product schedule will be maintained and that the product will be delivered on time.
- Support :degree of uncertainty that the resultant software will be easy to correct ,adapt ,and enhance. "How easy will it be to maintain and update?"
- It is required that the project manager identify the risk drivers that affect software risk components.

Components Category		Performance	Support	Cost	Schedule
Catastrophic	1	Failure to meet the requirement would result in mission failure		Failure results in increased costs and schedule delays with expected values in excess of \$500K	
	2	Significant degradation to nonachievement of technical performance	Nonresponsive or unsupportable software	Significant financial shortages, budget overrun likely	Unachievable IOC
Critical	1	Failure to meet the requirement would degrade system performance to a point where mission success is questionable		Failure results in operational delays and/or increased costs with expected value of \$100K to \$500K	
	2	Some reduction in technical performance	Minor delays in software modifications	Some shortage of financial resources, possible overruns	Possible slippage in IOC
Marginal	1	Failure to meet the requirement would result in degradation of secondary mission		Costs, impacts, and/or recoverable schedule slips with expected value of \$1K to \$100K	
	2	Minimal to small reduction in technical performance	Responsive software support	Sufficient financial resources	Realistic, achievable schedule
Negligible	1	Failure to meet the requirement would create inconvenience or nonoperational impact		Error results in minor cost and/or schedule impact with expected value of less than \$1K	
	2	No reduction in technical performance	Easily supportable software	Possible budget underrun	Early achievable IOC

1. Catastrophic

Performance: Mission fails or major technical failure.

Support: Software may not work at all.

Cost: Very large extra cost (more than \$500K).

Schedule: Planned date cannot be met.

2. Critical

Performance: System performance badly degraded; mission success uncertain.

Support: Delays in support.

Cost: Big cost increase (\$100K–\$500K).

Schedule: Serious delays.

3. Marginal

Performance: Only secondary functions or performance slightly affected.

Support: Support is okay or slightly delayed.

Cost: Small to medium overruns (\$1K–\$100K).

Schedule: Some slippage, but still achievable.

4. Negligible

Performance: No real performance loss; just inconvenience.

Support: Software easy to support.

Cost: Very small or even reduced cost (under \$1K).

Schedule: Schedule is fine or even earlier than planned.

Which level (catastrophic, critical, marginal, negligible) do you think is most important to watch closely in a real project, and why?

Project risk assessment matrix

		Likelihood				
		Very unlikely [1]	Unlikely [2]	Possible [3]	Probable [4]	Very likely [5]
Impact	100% Catastrophic [5] 80%	5	10	15	20	25
	Significant [4] 60%	4	8	12	16	20
	Moderate [3] 40%	3	6	9	12	15
	Low [2] 20%	2	4	6	8	10
	0% Negligible [1]	1	2	3	4	5

Risk projection (risk estimation)

- Attempts to rate each risk in two ways:
 - i. Probability that the risk is real.
 - ii. Consequences of the problems associated with the risk.
- There are four risk projection steps intended to consider risks in a manner that leads to prioritization.
- Estimate a scale that reflects the perceived probability of a risk (e.g., catastrophic to negligible)
- Delineate the consequences of the risk (Understand the specific effects of a risk on software components, project timelines, or budget)
Estimate the impact of the risk on project and product (Quantify the overall effect on project goals (time, cost, quality)).
- Assess the overall accuracy of the risk projection (Document the confidence in the projection to avoid misinterpretations)

- A risk table provides simple techniques for risk projection.
- Begin by listing all the risks in the table with the help of risk item checklist.
- Each risk is categorized in the second table.
- Probability of occurrence of each risk is entered in the next column.
- Next, the impact of each risk is assessed.

Risk Table

Risks	Category	Probability	Impact	RMMM
Size estimate may be significantly low	PS	60%	2	
Larger number of users than planned	PS	30%	3	
Less reuse than planned	PS	70%	2	
End-users resist system	BU	40%	3	
Delivery deadline will be tightened	BU	50%	2	
Funding will be lost	CU	40%	1	
Customer will change requirements	PS	80%	2	
Technology will not meet expectations	TE	30%	1	
Lack of training on tools	DE	80%	3	
Staff inexperienced	ST	30%	2	
Staff turnover will be high	ST	60%	2	
•				
•				
•				

Impact values:

- 1—catastrophic
- 2—critical
- 3—marginal
- 4—negligible

Risk Category:

- PS = Project size risk
- BU = Business risk
- TE = Technology risk
- DE = Development risk
- ST = Stakeholder risk
- CU = Customer risk

Risk exposure

- $RE = P * C$
- P=probability of occurrence for each risk
- C=cost of project when risk occurs(impact)
- Risk Exposure can be computed for each risk ,once the estimation of the cost of the risk is made.
- The total RE for all the risks can provide a mean for adjusting the final cost.
- RE can be used to predict the probable increase in staff resources required at various points during the project schedule.

Risk exposure

For example, assume that software team defines a project risk in as follows:

- **Risk Identification:** Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.
- **Risk Probability:** 80% (likely)
- **Risk Impact:** 60 reusable software components were planned. If only 70 percent can be used, **18 components** would have to be developed from scratch (in addition to other custom software that has been scheduled for development). Since the average component is **100 LOC** and local data indicate that the software engineering cost for each LOC is **\$14.00**

Risk exposure

Solution:

- Overall cost (impact) to develop the components would be $18 \times 100 \times 14$
 $= \$25,200$.
- Risk Exposure. $RE = 0.80 \times 25,200$
 $\sim \$20,160$

Risk refinement

- During early stages of project planning, a risk may be stated quite generally. As time passes and more is learned about risk ,it may be possible to refine the risk into a set of more detailed risks each somewhere easy to monitor and manage.
- One way to do this is to represent the risk in Condition Transition Consequence format:

(A general risk is refined into smaller risks using:)

Condition → Transition → Consequence

Example:

Condition: Using new technology

Transition: Team has no experience

Consequence: Project delay

- Refinement helps to isolate the underlying risks and lead to easier analysis and response

RISK MITIGATION, MONITORING,MANAGEMENT(RMMM)

- An effective risk strategy must consider three issues:
- **Risk avoidance** (Steps taken *before* a risk occurs to prevent or reduce its impact (e.g., using proven technology instead of experimental ones).
- **Risk monitoring** (Continuously watching project progress and risk indicators to see if risks are materializing)
- **Risk management** (Actions taken *after* a risk happens to handle it (contingency planning))

Risk mitigation

- If a software team adopts a proactive approach to risk, avoidance is best strategy, achieved by developing a plan for mitigation.
- For ex, assume that high staff turnover is noted as a project risk r_1 . Based on past history and management intuition, the likelihood l_1 of high turnover is estimated to be 0.70 and the impact x_1 is produced as critical. i.e. high turnover will have a critical impact on project cost and schedule.
- To mitigate this risk, you would develop a strategy for reducing turnover

Risk mitigation strategies

Four Types of Risk Mitigation



Risk mitigation

- Here are key strategies with details:

Risk Acceptance: Choosing to accept the potential consequences, often for minor risks or when mitigation costs exceed benefits.

Example: Accepting minor budget overruns on a small project.

- **Risk Avoidance:** This involves stopping the activity or changing plans to completely eliminate exposure to a particular risk.

Example: Deciding not to launch a product in a high-risk market to avoid potential legal issues.

- **Risk Reduction (Mitigation Control):** Proactively implementing measures to decrease the probability or impact of a risk.

Example: Installing firewalls and antivirus software (layered security) to prevent cyberattacks.

- **Risk Transfer:** Shifting the financial burden or responsibility of a risk to a third party.

Example: Buying insurance for property damage or outsourcing a risky IT function to a specialized vendor.

Risk monitoring

- After risk mitigation ,risk monitoring activity commence. The project manager monitors the factors that may provide an indication of whether the risk is becoming more or less likely.
- e.g. In case of high staff turnover, the general attitude of team members based on project pressures ,interpersonal relationships among team members, potential problems with compensation and benefits, availability of jobs within the company and outside the company is monitored
- Project manager should also monitor the effectiveness of risk mitigation steps.
- Project manager should monitor work products carefully to ensure that each can stand on its own.

Risk management

Risk management and contingency planning assumes that mitigation efforts have failed and the risk has become a reality.

Example: the project is under way and number of people announce that they are leaving if the mitigation step is followed, backup is available, information is documented and knowledge has been dispersed across the team.

In addition temporarily focus resource and schedule(to readjust)to those function that are fully staffed, enabling new comers who must be added to team to get up speed.

Risk management

The example illustrates a two-part response when key team members leave unexpectedly:

- Immediate Action:** Use pre-planned backups, documented information, and cross-trained team members to cover the immediate gaps.
- Strategic Adjustment:** Temporarily shift focus and resources to the stable parts of the project, creating space and time for new team members to be hired and properly integrated (get "up to speed").

Risk information sheet

Risk ID: P02-4-32

Date: 5/9/09

Prob: 80%

Impact: high

Description:

Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.

Refinement/context:

Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards.

Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components.

Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.

Mitigation/monitoring:

1. Contact third party to determine conformance with design standards.
2. Press for interface standards completion; consider component structure when deciding on interface protocol.
3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.

Management/contingency plan/trigger:

RE computed to be \$20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly.

Trigger: Mitigation steps unproductive as of 7/1/09.

Current status:

5/12/09: Mitigation steps initiated.

Originator: D. Gagne

Assigned: B. Laster

This is a risk information sheet for a software project.

Risk: Only 70% of planned reusable components may actually be usable; the rest must be custom-built.

Probability: 80%, Impact: high.

Why this might happen (refinement/context):

Some components were built by a third party and may not follow the team's design standards.

Interface standards are not yet fixed, so components may not fit.

Some components use a programming language not supported on the target system.

Mitigation / monitoring (what to do now):

Talk to the third party and check if their components follow the design standards.

Push to finish interface standards; consider components while defining protocols.

Count how many components are in unsupported languages and see if language support can be added.

Management / contingency plan (if mitigation fails):

Expected extra cost due to this risk: \$20,200.

Plan project again assuming 18 components must be custom developed and assign staff for this.

Trigger: If by 7/1/09 the mitigation steps are not working, activate this contingency plan.

Current status (as of 5/12/09): Mitigation actions have started.

This sheet answers four things: *what the risk is, why it may happen, how we try to prevent it, and what we'll do if prevention fails.*